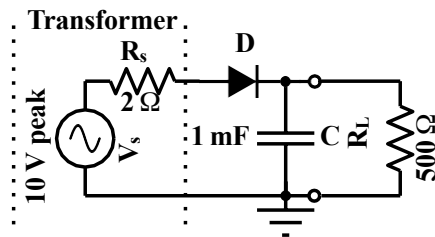


Tutorial-6

(AC to DC conversion and Introduction to Operational Amplifiers)

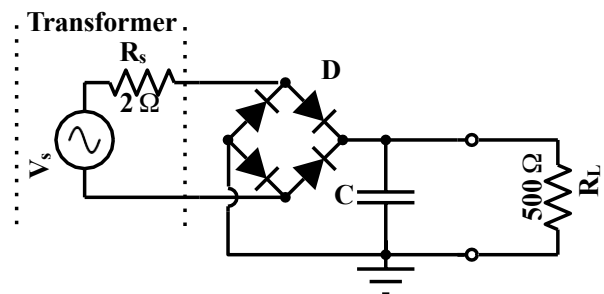
Q-1 A step down transformer generates an AC voltage of 10 V peak from 220 V, 50 Hz AC mains. The resistance of secondary winding is $2\ \Omega$. A diode rectifies the stepped down AC voltage and the half wave rectified AC is smoothed using a 1 mF capacitor. The load can be represented by a resistor of $500\ \Omega$. The resulting equivalent circuit is shown below. We'll assume the diode to be ideal in this problem.



During the positive half cycle, the diode conducts only when the AC voltage exceeds the voltage stored on the capacitor. Once the AC voltage drops below the capacitor voltage, the diode becomes an open circuit. In this state, the capacitor discharges through the load resistor.

- What is the time constant during the charge phase? (Replace the diode by a short circuit and the AC voltage source along with R_s and R_L to its Thevenin equivalent).
- What is the time constant during discharge? (Diode is replaced by an open circuit).
- Since the discharge time constant is much longer, we make the simplifying assumption that the discharge duration is nearly the whole AC cycle of 20 ms. Assume that the capacitor charges up to the Thevenin voltage during the short charging cycle. To what voltage will the capacitor discharge before the charge cycle begins again?
- What is the peak to peak ripple voltage?

Q-2 You are tasked with designing a Full-Wave Bridge Rectifier circuit with a capacitor filter to supply a steady 10 V DC to a $500\ \Omega$ load. The circuit is powered by a standard 50 Hz AC transformer. The output ripple voltage $V_{r(pp)}$ must not exceed 0.2 V. Assume standard silicon diodes with a forward drop of 0.7 V.



- Calculate the minimum capacitance (C) required to meet the ripple specification.
- Determine the required Peak Secondary Voltage (V_m) and the corresponding RMS Voltage rating required from the transformer to maintain the 10 V DC output.

Q-3 You are designing a 5.0 V regulated power supply and evaluating two possible circuit configurations, A) A single Zener Diode in Reverse Breakdown, B) A series string of 7 standard Silicon Diodes in Forward Bias.

You decide to compare the two designs using the manufacturer's datasheet values below.

A) Zener Diode (Reverse Biased): Voltage at low current (10 mA): $V_Z = 4.95\text{ V}$ and Voltage at high current (100 mA): $V_Z = 5.05\text{ V}$,

B) Standard Diode (Forward Biased): Voltage at low current (10 mA): $V_F = 0.70\text{ V}$ and Voltage at high current (100 mA): $V_F = 0.85\text{ V}$.

- Calculate the change in total output voltage (i.e. voltage regulation) for both designs when the current increases from 10 mA to 100 mA.
- Based on your calculation in part (a), determine which design is better for a voltage regulator. Explain your choice by comparing the Dynamic Resistance (slope of the V-I curve) of the two designs.

Q-4 You are designing an interface card for a vibration sensor which outputs an AC sine wave that swings between -5 V and $+5\text{ V}$. The microcontroller has an analog input that can only safely handle positive voltages (0 V to 10 V). If the signal goes negative, the chip will be damaged. To fix this, you design a interface circuit (using a capacitor and a diode) between the sensor and the microcontroller. Given: Sensor Frequency (f): 1 kHz , Microcontroller Input Resistance (R): $10\text{ k}\Omega$. Assume the Diode is Ideal (acts as a perfect switch with 0 V drop when ON).

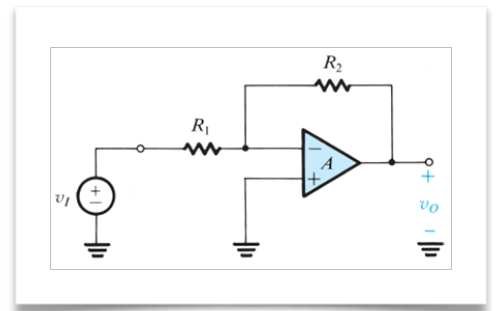
Show the design of the interface circuit, and its analysis which results in desired output. Discuss the choice of the value of the capacitor.

Q-5 In your design of a precision amplifier circuit for a medical device, you have chosen an inverting amplifier configuration with resistor values $R_1 = 1\text{ k}\Omega$ and $R_2 = 10\text{ k}\Omega$.

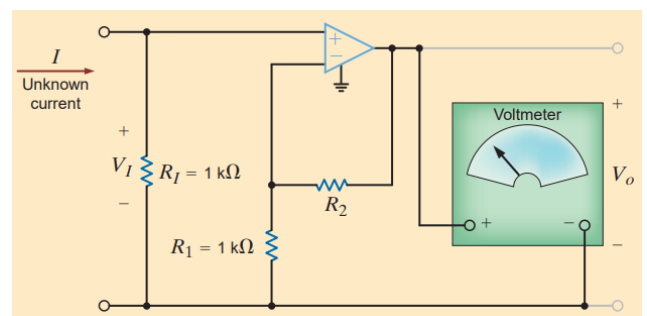
Your purchasing manager has offered you two Op-Amp chips to choose from: A) a Gold standard chip with an Open-Loop Gain of $A = 100,000$ but costs 1000 Rupees and, B) a Budget chip with an Open-Loop Gain of $A = 1000$, but cost 30 Rupees.

To make your decision, you are advised to

- First calculate the ideal closed-loop gain assuming $A = \infty$.
- Then, determine the actual closed-loop gain for both the Gold and Budget chips.
- Finally, calculate the percentage error of the Budget chip compared to the ideal target and determine if the cheaper chip is sufficient for a design requiring a 1% tolerance.



Q-6 You are given a circuit to design an electronic ammeter circuit that measures an unknown current, I , by first converting it into a voltage and then amplifying it for a display. The input current flows through a shunt resistor, $R_1 = 1\text{ k}\Omega$, to ground, creating a small sensing voltage (V_I). This voltage is then fed into a non-inverting Op-Amp configuration defined by an input resistor $R_1 = 1\text{ k}\Omega$ and a feedback resistor R_2 .

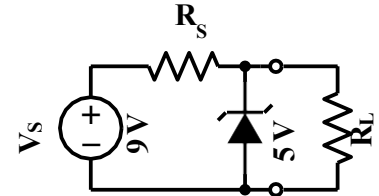


The goal is to calibrate the meter so that the output voltage (V_{out}) is exactly 10 V for every 1 mA of input current. Determine the value of the feedback resistor, R_2 , required to achieve this specific sensitivity. [Assume the Open Loop Gain of OpAMP to be sufficiently high i.e. $100,000$]

Additional (Optional) questions

Q- We want to design a regulated supply using an avalanche diode.

The load requires a regulated supply of 5.0 V. Current drawn by the load may vary over the range 100 mA $\pm$ 10 mA depending on its mode of operation. An avalanche diode with a breakdown voltage of 5.0 V is available.



- a) Initially, let us assume that the drop across the avalanche diode remains fixed at 5.0 V provided the current through it is > 1 mA in the breakdown region. Input to the The regulator circuit is provided by an unregulated DC source whose output voltage is 9 V.

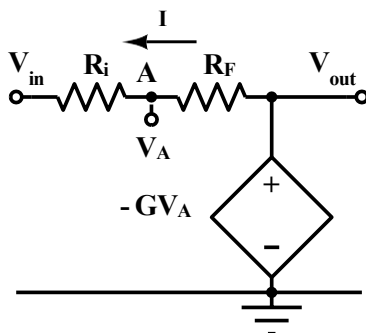
Choose a value of R_s such that the Avalanche diode remains in the breakdown region with current through it > 1 mA for any value of the current drawn by the load in its specified range (90 mA to 110 mA).

- b) With the value of R_s chosen as in part a) above, what is the current through the avalanche diode when the load current is at the minimum value of its range ($= 90$ mA).
- c) What is the worst case power dissipation through R_s and the avalanche diode?
- d) Now assume that the output voltage of the unregulated supply may vary over the range 9 ± 1 V. What should be the value for R_s now and for this value, what is the worst case dissipation in R_s and the diode?
- e) Assume that the voltage across the diode is not fixed at 5 V, but increases by 1 mV for every mA of current through the diode. This model applies only when the diode is in the break down region. In this case, the diode can be modeled as a voltage source of 5 V in series with a 1Ω resistor.

V_s can vary over the range 9 ± 1 V, the load current can vary over the range 100 ± 10 mA and $R_s = 27 \Omega$

- i) Line regulation for a power supply is defined as $\Delta V_{out}/\Delta V_{in}$ at constant load current. What is the line regulation for this supply with the chosen value of R_s and load current of 100 mA?
- ii) Load regulation for a power supply is defined as $\Delta V_{out}/\Delta I_L$ at constant input Voltage. What is the load regulation for this power supply with the input voltage $= 9$ V?
- iii) What is the worst case power dissipation in R_s and the avalanche diode?

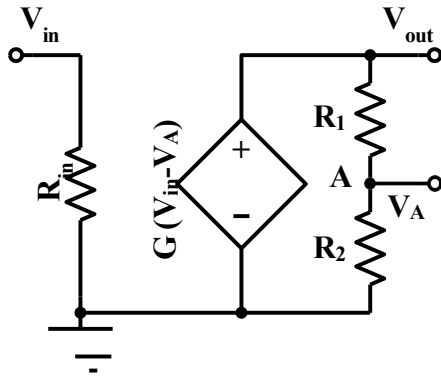
Q- Consider the circuit below, which uses a voltage controlled voltage source (VCVS).



The output of the voltage controlled voltage source is proportional to the voltage at A with proportionality constant G . (G is known as the “open loop gain”).

- Find V_{out}/V_{in} as a function of R_i , R_F and G
- What is output voltage V_{out} as $G \rightarrow \infty$?
- What is the voltage at A (V_A) as $G \rightarrow \infty$?

Q- Now consider the circuit below which also uses a voltage controlled voltage source.



The output of the voltage controlled voltage source is proportional to the *difference* between the input voltage V_{in} and the voltage at node A (V_A) with proportionality constant G .

- Find V_{out} as a function of V_{in} and G
- What is the voltage at the output (V_{out}) as $G \rightarrow \infty$?
- What is the voltage at A (V_A) as $G \rightarrow \infty$?